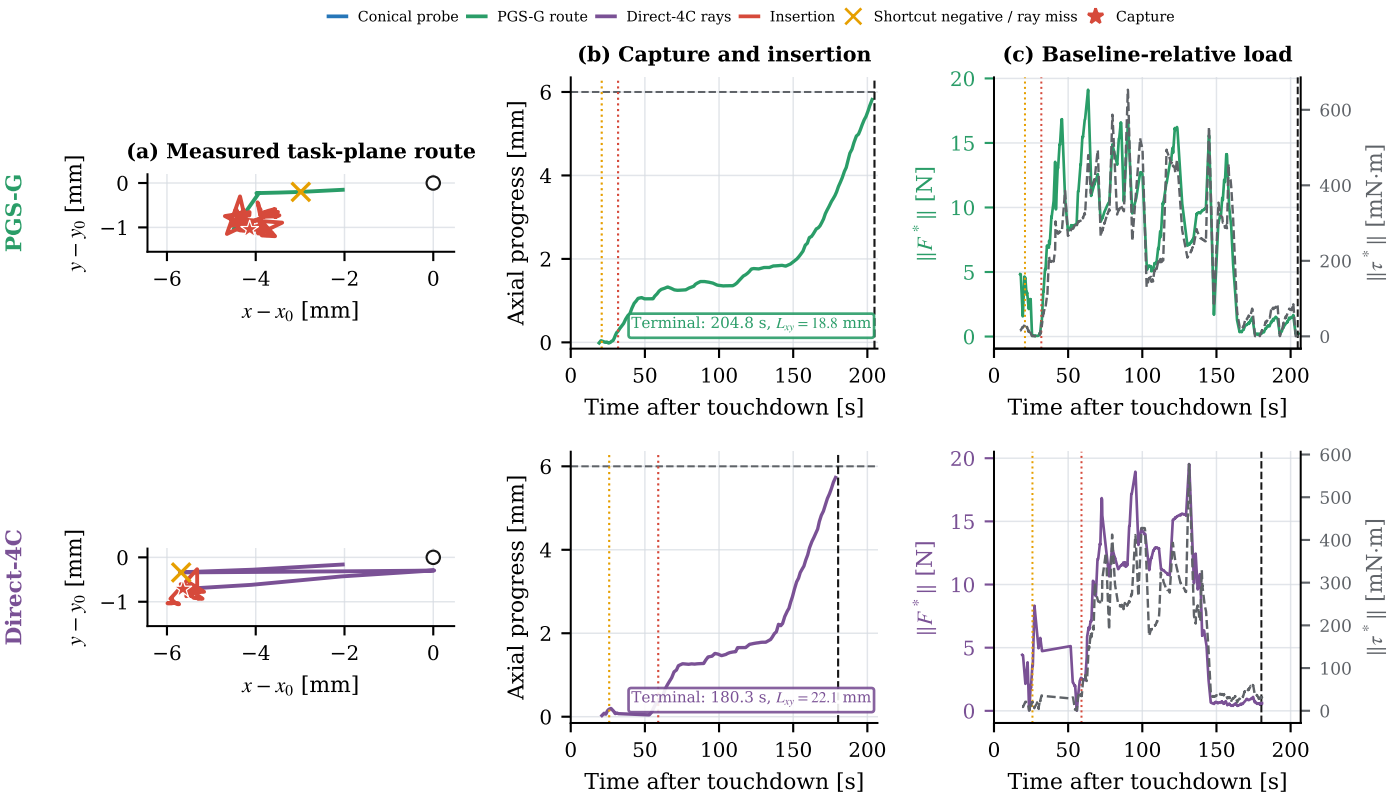

Conical probe
PGS-G route
Direct-4C rays
Insertion
Shortcut negative / ray miss
Capture
PGS-G
Direct-4C
(a) Measured task-plane route
(b) Capture and insertion
(c) Baseline-relative load
$y - y_0$ [mm]
$x - x_0$ [mm]
Axial progress [mm]
Time after touchdown [s]
Terminal: 204.8 s, $L_{xy} = 18.8$ mm
Terminal: 180.3 s, $L_{xy} = 22.1$ mm
$\|F^*\|$ [N]
$\|\tau^*\|$ [mN·m]